

Titanium Grade-5 Ti6Al4V

TITANIUM GRADE-5 TI-6AL-4V

Common Name	
Ti-6Al-4V	Titanium Grade 5
Ti-6-4	Ti64
UNS	
R56400	
General Information	
Ti-6Al-4V alloy is the most widely used titanium alloy of the alpha-plus-beta class, and is also the most common of all titanium alloys. The alloy is castable and is utilized "as cast" in sporting goods. The wrought material is used in aerospace, medical, and other applications where moderate strength, good strength to weight, and favorable corrosion properties are required. The alloy is available as castings, wire, bar, plate, sheet, forgings, rings, and billet.	
Common Specification	
Specification	Product Form
AMS 4911	Strip, Sheet, and Plate, Annealed
AMS 4920	Forgings, Alpha-Beta or Beta Processed, Annealed
AMS 4928	Bar, Wire, Forgings, Ring, Annealed
AMS 4965, AMS 4963 & AMS 4965, AMS 4963 & AMS 4967 (Capable of)	Bar, Wire, Forgings, Ring, Solution Treated & Aged
AMS-T-9047	-
ASTM B348 (Grade 5)	Bar and Billet, Annealed
ASTM B367 (Grade 5)	Castings
ASTM F1472	Wrought Alloy for Surgical Implants
AWS A5.16 (ERTi-5)	Weld Wire

Chemistry Requirements								
N	C	H	Fe	O	Al	V	Y	Ti
0.05	0.08	0.125	0.4	0.2	5.5-6.75	3.5-4.5	0.005	Balance

Note: Chemical requirements are not consistent between specifications. Check referenced specifications.

Minimum Tensile Properties				
Condition	UTS ksi (Mpa)	>0.2%YS ksi (MPA)	% El.	% RA*
Solution Treated & Aged	170 (1172)	160 (1103)	8-10	15-20

Note: Mechanical properties vary with diameter. Check referenced specifications.

Typical Tensile Properties				
Condition	UTS ksi (Mpa)	>0.2%YS ksi (MPA)	% El.	% RA*
Annealed	160 (1110)	148 (1020)	15	–
Solution Treated & Aged (BSTA)	174 (1200)	152 (1050)	7	–
Solution Treated & Aged (STA-1)	176 (1210)	163 (1120)	13	–
Solution Treated & Aged (STA-2)	229 (1580)	204 (1410)	4	–

Note: Typical properties are not to be utilized as a requirement, but are only listed for guidance. These properties may or may not be attainable in all circumstances.

* %Ra not required by all specifications

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